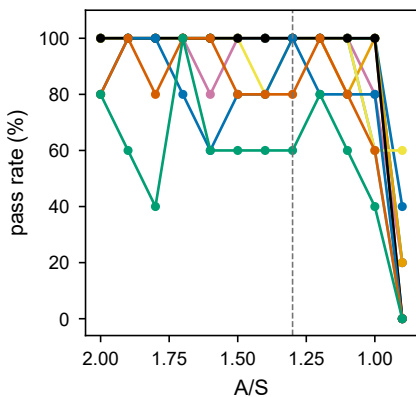
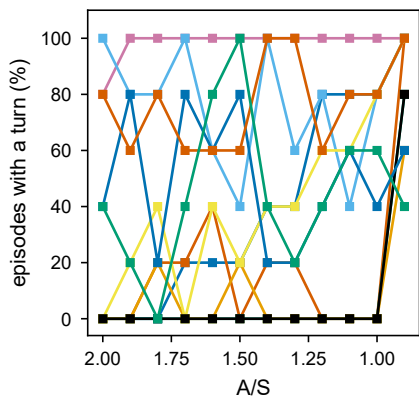


# The eleven models together

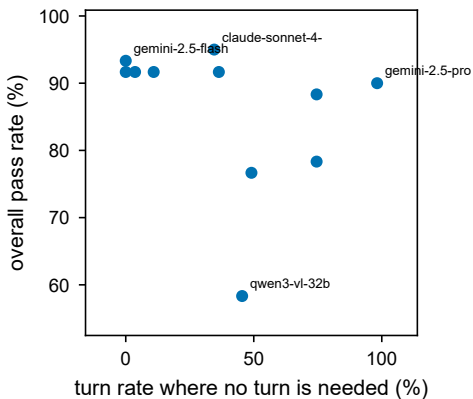
pass rate per aperture



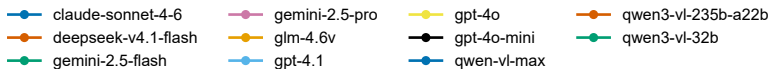
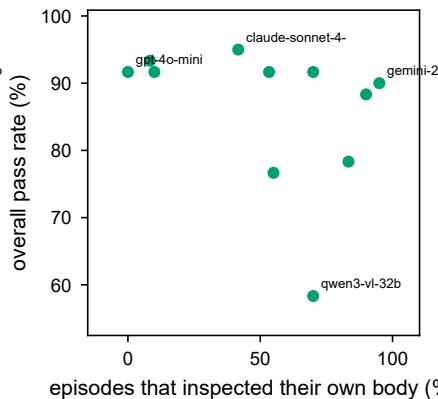
turn rate per aperture (flat = a fixed policy)



spurious rotation vs success



self-inspection vs success



Every model, sorted by overall pass rate  
 pass rate at the two apertures that need rotation (A/S 1.0, 0.9); wide turn% = turning where none

| model            | pass% | narrow | turn% | wide<br>turn% | first<br>turn x | look% | pass <br>look | pass <br>nolook |
|------------------|-------|--------|-------|---------------|-----------------|-------|---------------|-----------------|
| claude-sonnet-4- | 95    | 7/10   | 40    | 35            | 2.31            | 42    | 88            | 100             |
| gemini-2.5-flash | 93    | 6/10   | 7     | 0             | 7.06            | 8     | 60            | 96              |
| deepseek-v4.1-fl | 92    | 5/10   | 18    | 11            | 3.77            | 53    | 84            | 100             |
| glm-4.6v         | 92    | 6/10   | 8     | 4             | 5.00            | 10    | 50            | 96              |
| gpt-4o           | 92    | 6/10   | 42    | 36            | 1.64            | 70    | 88            | 100             |
| gpt-4o-mini      | 92    | 5/10   | 7     | 0             | 7.25            | 0     | –             | 92              |
| gemini-2.5-pro   | 90    | 5/10   | 98    | 98            | 1.99            | 95    | 89            | 100             |
| gpt-4.1          | 88    | 3/10   | 77    | 75            | 1.71            | 90    | 87            | 100             |
| qwen3-vl-235b-a2 | 78    | 3/10   | 77    | 75            | 1.82            | 83    | 76            | 90              |
| qwen-vl-max      | 77    | 4/10   | 50    | 49            | 2.52            | 55    | 58            | 100             |
| qwen3-vl-32b     | 58    | 2/10   | 45    | 45            | 1.22            | 70    | 48            | 83              |

Pass rate spans 58-95%, and the two apertures that need rotation already separate the models (claude-sonnet- 7/10, gemini-2.5-fla 6/10, deepseek-v4.1- 5/10).

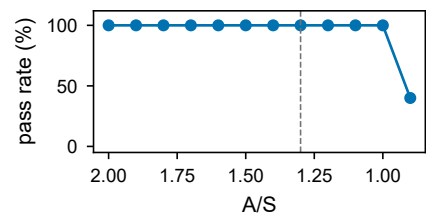
Turning where no turn is needed averages 39% and ranges from 0% (gemini-2.5-fla) to 98% (gemini-2.5-pro), so the models differ by more than an order of magnitude on a decision the aperture does not require at all.

Inspecting its own body, compared inside each aperture rather than pooled (pooling would be confounded, because looking down clusters at the narrow Levels): -9.1 percentage points on average, positive in 1 of 11 models, so inspection does not buy success by itself.

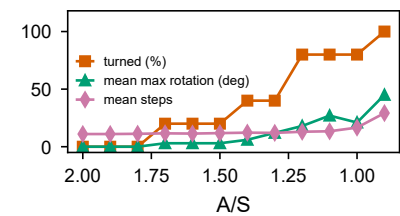
The reasoning names the opening, the turn and the model's own width; what fails is the amount of rotation and the alignment, not knowing that a turn is needed.

claude-sonnet-4-6 — 60 episodes, overall pass 95%

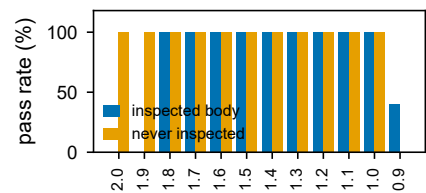
outcome vs aperture



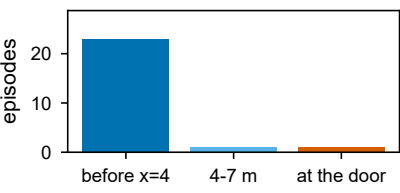
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.9 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.9 | 1.083 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.8 | 1.026 | 5/5  | 0     | 0     | 11.2  | 0      | 0.2    | 20    | 100       | 100         |
| 1.7 | 0.969 | 5/5  | 20    | 0     | 11.6  | 3      | 0.6    | 20    | 100       | 100         |
| 1.6 | 0.912 | 5/5  | 20    | 0     | 11.4  | 3      | 0.4    | 20    | 100       | 100         |
| 1.5 | 0.855 | 5/5  | 20    | 0     | 11.8  | 3      | 0.8    | 60    | 100       | 100         |
| 1.4 | 0.798 | 5/5  | 40    | 0     | 12.2  | 6      | 1.2    | 60    | 100       | 100         |
| 1.3 | 0.741 | 5/5  | 40    | 20    | 12.0  | 12     | 1.0    | 20    | 100       | 100         |
| 1.2 | 0.684 | 5/5  | 80    | 0     | 13.0  | 18     | 2.0    | 60    | 100       | 100         |
| 1.1 | 0.627 | 5/5  | 80    | 40    | 13.4  | 27     | 2.4    | 60    | 100       | 100         |
| 1.0 | 0.570 | 5/5  | 80    | 0     | 16.6  | 21     | 5.6    | 80    | 100       | 100         |
| 0.9 | 0.513 | 2/5  | 100   | 100   | 29.2  | 45     | 14.2   | 100   | 40        | —           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): opening 100%, align 100%, turn 92%, width 70%, sideways 40%, blocked 28%, budget 3%

Actions: forward 84.9%, turn\_left 5.2%, look\_down 3.8%, turn\_right 2.7%, right 1.8%, left 1.3%

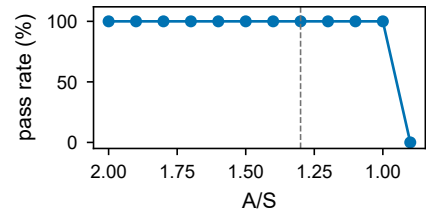
Strategy labels: FRONTAL 60%, ROT\_EARLY 35%, ROT\_MID 3%, ROT\_LATE 2%

It turned in 40% of episodes (first turn at x = 2.3 m on average), and in 35% of the episodes at apertures that need no turn at all; its passes waste 2.4 steps over the optimum on average.

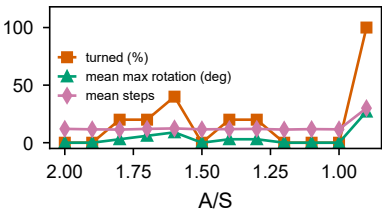
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 9 apertures positive).

deepseek-v4.1-flash — 60 episodes, overall pass 92%

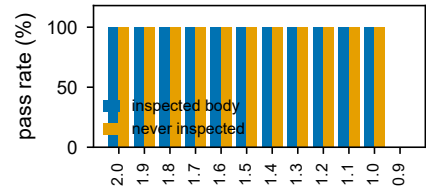
outcome vs aperture



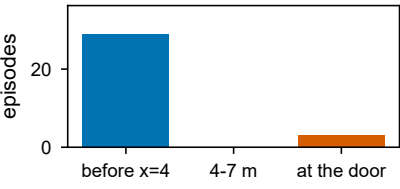
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.6 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 0     | 0     | 12.0  | 0      | 1.0    | 80    | 100       | 100         |
| 1.9 | 1.083 | 5/5  | 0     | 0     | 11.6  | 0      | 0.6    | 60    | 100       | 100         |
| 1.8 | 1.026 | 5/5  | 20    | 0     | 11.4  | 3      | 0.4    | 20    | 100       | 100         |
| 1.7 | 0.969 | 5/5  | 20    | 0     | 12.0  | 6      | 1.0    | 60    | 100       | 100         |
| 1.6 | 0.912 | 5/5  | 40    | 0     | 12.4  | 9      | 1.4    | 60    | 100       | 100         |
| 1.5 | 0.855 | 5/5  | 0     | 0     | 11.6  | 0      | 0.6    | 60    | 100       | 100         |
| 1.4 | 0.798 | 5/5  | 20    | 0     | 11.8  | 3      | 0.8    | 40    | 100       | 100         |
| 1.3 | 0.741 | 5/5  | 20    | 0     | 12.0  | 3      | 1.0    | 20    | 100       | 100         |
| 1.2 | 0.684 | 5/5  | 0     | 0     | 11.4  | 0      | 0.4    | 40    | 100       | 100         |
| 1.1 | 0.627 | 5/5  | 0     | 0     | 11.8  | 0      | 0.8    | 60    | 100       | 100         |
| 1.0 | 0.570 | 5/5  | 0     | 0     | 11.6  | 0      | 0.6    | 40    | 100       | 100         |
| 0.9 | 0.513 | 0/5  | 100   | 0     | 30.0  | 27     | 15.0   | 100   | 0         | -           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): opening 100%, align 100%, width 80%, turn 43%, blocked 32%, sideways 10%, budget 3%

Actions: forward 86.1%, look\_down 5.1%, turn\_left 3.0%, turn\_right 2.0%, right 1.6%, left 1.6%

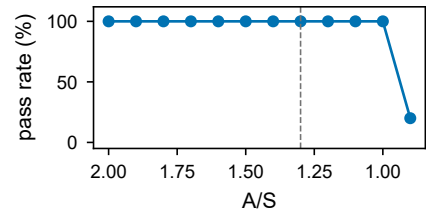
Strategy labels: FRONTAL 82%, ROT\_EARLY 10%, ROT\_LATE 8%

It turned in 18% of episodes (first turn at x = 3.8 m on average), and in 11% of the episodes at apertures that need no turn at all; its passes waste 2.0 steps over the optimum on average.

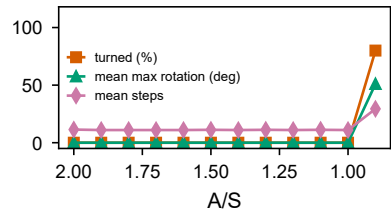
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 11 apertures positive).

gemini-2.5-flash — 60 episodes, overall pass 93%

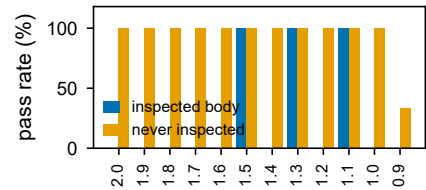
outcome vs aperture



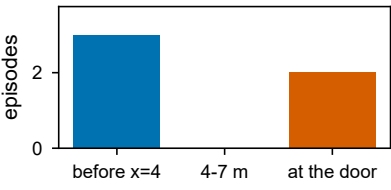
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 4.0 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 0     | 0     | 11.4  | 0      | 0.4    | 0     | —         | 100         |
| 1.9 | 1.083 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.8 | 1.026 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.7 | 0.969 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.6 | 0.912 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.5 | 0.855 | 5/5  | 0     | 0     | 11.2  | 0      | 0.2    | 20    | 100       | 100         |
| 1.4 | 0.798 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.3 | 0.741 | 5/5  | 0     | 0     | 11.2  | 0      | 0.2    | 20    | 100       | 100         |
| 1.2 | 0.684 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.1 | 0.627 | 5/5  | 0     | 0     | 11.2  | 0      | 0.2    | 20    | 100       | 100         |
| 1.0 | 0.570 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 0.9 | 0.513 | 1/5  | 80    | 100   | 29.4  | 51     | 14.4   | 40    | 0         | 33          |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): opening 100%, align 92%, turn 32%, blocked 22%, width 12%, sideways 3%

Actions: forward 89.2%, turn\_right 2.5%, turn\_left 2.2%, look\_down 2.1%, left 1.5%, right 1.2%

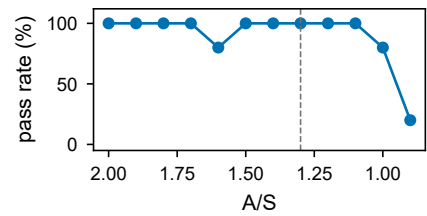
Strategy labels: FRONTAL 93%, ROT\_LATE 5%, ROT\_MID 2%

It turned in 7% of episodes (first turn at x = 7.1 m on average), and in 0% of the episodes at apertures that need no turn at all; its passes waste 1.3 steps over the optimum on average.

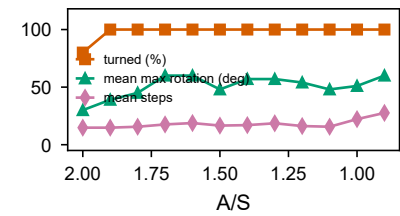
Inspecting its own body, compared inside each aperture: -8.3 percentage points on the pass rate (0 of 4 apertures positive).

gemini-2.5-pro — 60 episodes, overall pass 90%

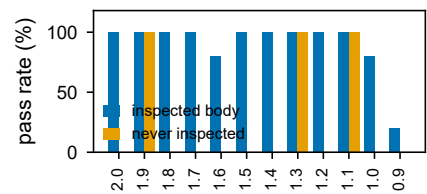
outcome vs aperture



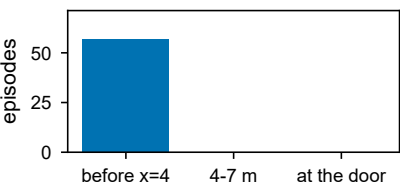
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.6 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 80    | 40    | 14.8  | 30     | 3.8    | 100   | 100       | —           |
| 1.9 | 1.083 | 5/5  | 100   | 40    | 14.8  | 39     | 3.8    | 80    | 100       | 100         |
| 1.8 | 1.026 | 5/5  | 100   | 60    | 15.6  | 45     | 4.6    | 100   | 100       | —           |
| 1.7 | 0.969 | 5/5  | 100   | 80    | 17.6  | 60     | 6.6    | 100   | 100       | —           |
| 1.6 | 0.912 | 4/5  | 100   | 75    | 18.8  | 60     | 7.8    | 100   | 80        | —           |
| 1.5 | 0.855 | 5/5  | 100   | 60    | 16.6  | 48     | 5.6    | 100   | 100       | —           |
| 1.4 | 0.798 | 5/5  | 100   | 60    | 17.0  | 57     | 6.0    | 100   | 100       | —           |
| 1.3 | 0.741 | 5/5  | 100   | 40    | 18.6  | 57     | 7.6    | 80    | 100       | 100         |
| 1.2 | 0.684 | 5/5  | 100   | 80    | 16.2  | 54     | 5.2    | 100   | 100       | —           |
| 1.1 | 0.627 | 5/5  | 100   | 40    | 15.6  | 48     | 4.6    | 80    | 100       | 100         |
| 1.0 | 0.570 | 4/5  | 100   | 50    | 22.2  | 51     | 11.2   | 100   | 80        | —           |
| 0.9 | 0.513 | 1/5  | 100   | 100   | 27.6  | 60     | 12.6   | 100   | 20        | —           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): opening 100%, width 100%, turn 98%, sideways 98%, align 95%, blocked 70%, budget 2%

Actions: forward 61.5%, turn\_right 20.9%, look\_down 8.8%, turn\_left 7.1%, right 0.8%, left 0.6%

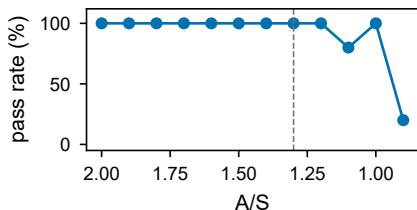
Strategy labels: ROT\_EARLY 98%, FRONTAL 2%

It turned in 98% of episodes (first turn at x = 2.0 m on average), and in 98% of the episodes at apertures that need no turn at all; its passes waste 6.6 steps over the optimum on average.

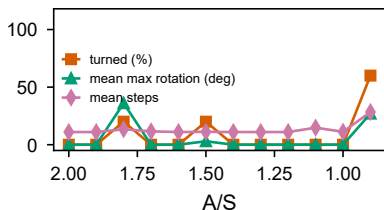
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 3 apertures positive).

# glm-4.6v — 60 episodes, overall pass 92%

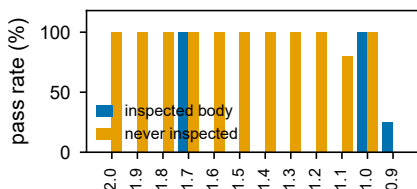
outcome vs aperture



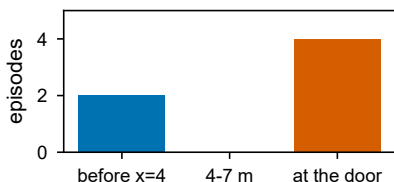
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 5.2 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.9 | 1.083 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.8 | 1.026 | 5/5  | 20    | 0     | 13.4  | 36     | 2.4    | 0     | —         | 100         |
| 1.7 | 0.969 | 5/5  | 0     | 0     | 11.6  | 0      | 0.6    | 20    | 100       | 100         |
| 1.6 | 0.912 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.5 | 0.855 | 5/5  | 20    | 0     | 11.2  | 3      | 0.2    | 0     | —         | 100         |
| 1.4 | 0.798 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.3 | 0.741 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.2 | 0.684 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.1 | 0.627 | 4/5  | 0     | 0     | 14.8  | 0      | 3.8    | 0     | —         | 80          |
| 1.0 | 0.570 | 5/5  | 0     | 0     | 11.2  | 0      | 0.2    | 20    | 100       | 100         |
| 0.9 | 0.513 | 1/5  | 60    | 100   | 28.4  | 27     | 13.4   | 80    | 25        | 0           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): align 65%, opening 43%, width 27%, blocked 13%, turn 12%, sideways 3%

Actions: forward 93.7%, turn\_right 2.3%, left 1.5%, turn\_left 1.1%, look\_down 0.8%, right 0.5%

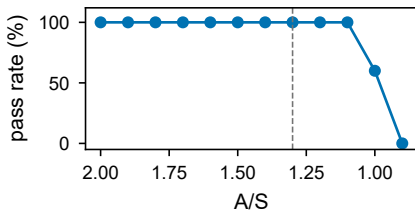
Strategy labels: FRONTAL 90%, ROT\_LATE 5%, ROT\_EARLY 3%, SIDEWAYS 2%

It turned in 8% of episodes (first turn at x = 5.0 m on average), and in 4% of the episodes at apertures that need no turn at all; its passes waste 1.7 steps over the optimum on average.

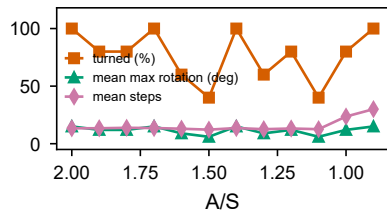
Inspecting its own body, compared inside each aperture: +8.3 percentage points on the pass rate (1 of 3 apertures positive).

# gpt-4.1 — 60 episodes, overall pass 88%

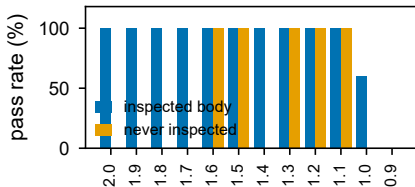
outcome vs aperture



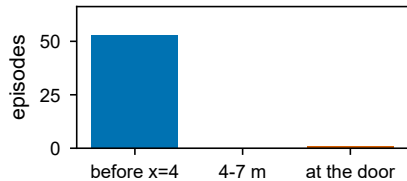
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.4 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 100   | 0     | 13.4  | 15     | 2.4    | 100   | 100       | —           |
| 1.9 | 1.083 | 5/5  | 80    | 0     | 13.4  | 12     | 2.4    | 100   | 100       | —           |
| 1.8 | 1.026 | 5/5  | 80    | 0     | 13.8  | 12     | 2.8    | 100   | 100       | —           |
| 1.7 | 0.969 | 5/5  | 100   | 0     | 13.6  | 15     | 2.6    | 100   | 100       | —           |
| 1.6 | 0.912 | 5/5  | 60    | 0     | 13.0  | 9      | 2.0    | 80    | 100       | 100         |
| 1.5 | 0.855 | 5/5  | 40    | 0     | 12.2  | 6      | 1.2    | 60    | 100       | 100         |
| 1.4 | 0.798 | 5/5  | 100   | 0     | 13.6  | 15     | 2.6    | 100   | 100       | —           |
| 1.3 | 0.741 | 5/5  | 60    | 0     | 12.6  | 9      | 1.6    | 80    | 100       | 100         |
| 1.2 | 0.684 | 5/5  | 80    | 0     | 13.2  | 12     | 2.2    | 80    | 100       | 100         |
| 1.1 | 0.627 | 5/5  | 40    | 0     | 12.6  | 6      | 1.6    | 80    | 100       | 100         |
| 1.0 | 0.570 | 3/5  | 80    | 0     | 23.6  | 12     | 12.6   | 100   | 60        | —           |
| 0.9 | 0.513 | 0/5  | 100   | 0     | 30.0  | 15     | 15.0   | 100   | 0         | —           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): opening 100%, align 98%, width 90%, turn 77%, blocked 23%, sideways 2%

Actions: forward 76.2%, look\_down 13.2%, turn\_left 5.1%, right 1.6%, left 1.3%, look\_left 1.3%

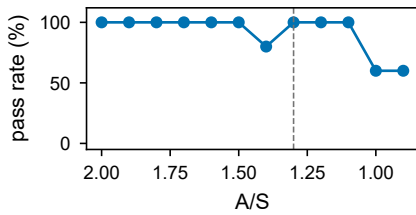
Strategy labels: ROT\_EARLY 70%, FRONTAL 23%, SIDEWAYS 7%

It turned in 77% of episodes (first turn at x = 1.7 m on average), and in 75% of the episodes at apertures that need no turn at all; its passes waste 4.1 steps over the optimum on average.

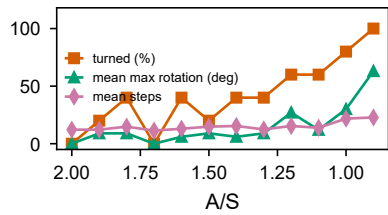
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 5 apertures positive).

# gpt-4o — 60 episodes, overall pass 92%

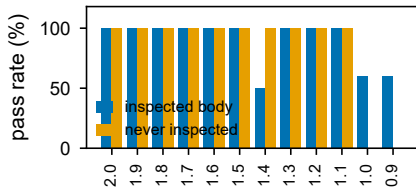
outcome vs aperture



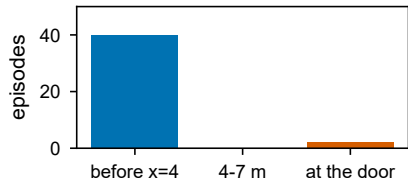
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.6 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 0     | 0     | 12.2  | 0      | 1.2    | 40    | 100       | 100         |
| 1.9 | 1.083 | 5/5  | 20    | 20    | 12.2  | 9      | 1.2    | 60    | 100       | 100         |
| 1.8 | 1.026 | 5/5  | 40    | 0     | 14.8  | 9      | 3.8    | 80    | 100       | 100         |
| 1.7 | 0.969 | 5/5  | 0     | 0     | 11.4  | 0      | 0.4    | 40    | 100       | 100         |
| 1.6 | 0.912 | 5/5  | 40    | 0     | 13.0  | 6      | 2.0    | 60    | 100       | 100         |
| 1.5 | 0.855 | 5/5  | 20    | 20    | 14.8  | 9      | 3.8    | 80    | 100       | 100         |
| 1.4 | 0.798 | 4/5  | 40    | 0     | 15.4  | 6      | 4.4    | 40    | 50        | 100         |
| 1.3 | 0.741 | 5/5  | 40    | 0     | 12.4  | 9      | 1.4    | 80    | 100       | 100         |
| 1.2 | 0.684 | 5/5  | 60    | 20    | 15.4  | 27     | 4.4    | 80    | 100       | 100         |
| 1.1 | 0.627 | 5/5  | 60    | 0     | 13.8  | 12     | 2.8    | 80    | 100       | 100         |
| 1.0 | 0.570 | 3/5  | 80    | 67    | 21.8  | 30     | 10.8   | 100   | 60        | -           |
| 0.9 | 0.513 | 3/5  | 100   | 100   | 22.8  | 63     | 7.8    | 100   | 60        | -           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): opening 88%, width 87%, align 85%, turn 45%, blocked 15%

Actions: forward 77.4%, look\_down 10.9%, turn\_left 6.8%, look\_left 1.7%, left 1.0%, look\_right 1.0%

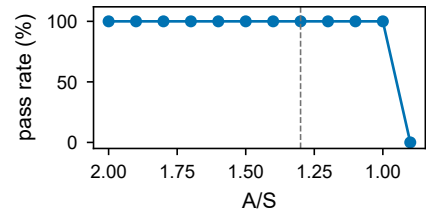
Strategy labels: FRONTAL 58%, ROT\_EARLY 38%, SIDEWAYS 3%

It turned in 42% of episodes (first turn at x = 1.6 m on average), and in 36% of the episodes at apertures that need no turn at all; its passes waste 3.7 steps over the optimum on average.

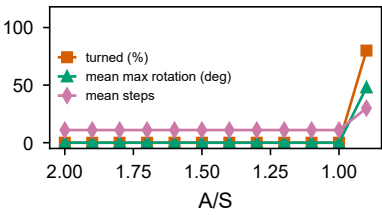
Inspecting its own body, compared inside each aperture: -5.0 percentage points on the pass rate (0 of 10 apertures positive).

gpt-4o-mini — 60 episodes, overall pass 92%

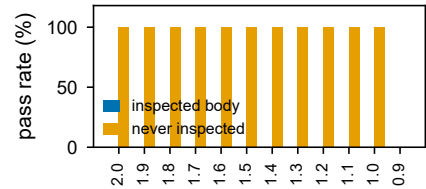
outcome vs aperture



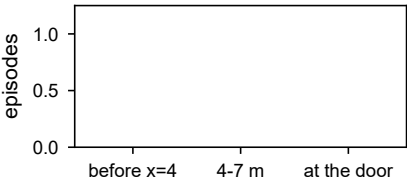
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 0.0 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.9 | 1.083 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.8 | 1.026 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.7 | 0.969 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.6 | 0.912 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.5 | 0.855 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.4 | 0.798 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.3 | 0.741 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.2 | 0.684 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.1 | 0.627 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 1.0 | 0.570 | 5/5  | 0     | 0     | 11.0  | 0      | 0.0    | 0     | —         | 100         |
| 0.9 | 0.513 | 0/5  | 80    | 0     | 30.0  | 48     | 15.0   | 0     | —         | 0           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): align 47%, blocked 8%, turn 7%

Actions: forward 89.7%, left 3.6%, right 2.8%, turn\_left 2.5%, turn\_right 1.3%, look\_left 0.1%

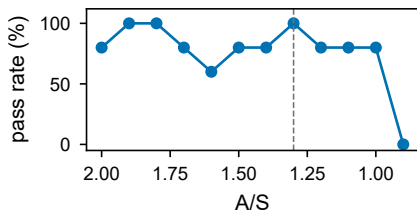
Strategy labels: FRONTAL 92%, ROT\_LATE 7%, SIDEWAYS 2%

It turned in 7% of episodes (first turn at x = 7.2 m on average), and in 0% of the episodes at apertures that need no turn at all; its passes waste 1.2 steps over the optimum on average.

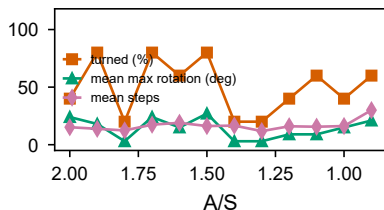
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 0 apertures positive).

qwen-vl-max — 60 episodes, overall pass 77%

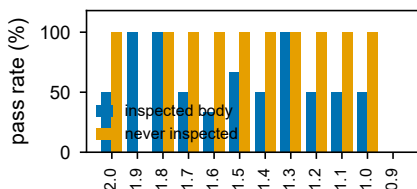
outcome vs aperture



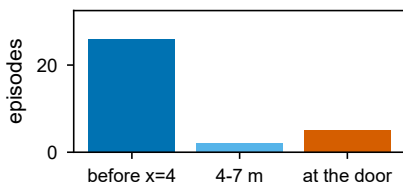
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 2.7 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 4/5  | 40    | 0     | 15.2  | 24     | 4.2    | 40    | 50        | 100         |
| 1.9 | 1.083 | 5/5  | 80    | 0     | 13.8  | 18     | 2.8    | 100   | 100       | —           |
| 1.8 | 1.026 | 5/5  | 20    | 0     | 12.4  | 3      | 1.4    | 80    | 100       | 100         |
| 1.7 | 0.969 | 4/5  | 80    | 0     | 17.4  | 24     | 6.4    | 40    | 50        | 100         |
| 1.6 | 0.912 | 3/5  | 60    | 0     | 19.0  | 15     | 8.0    | 60    | 33        | 100         |
| 1.5 | 0.855 | 4/5  | 80    | 0     | 16.2  | 27     | 5.2    | 60    | 67        | 100         |
| 1.4 | 0.798 | 4/5  | 20    | 0     | 16.6  | 3      | 5.6    | 40    | 50        | 100         |
| 1.3 | 0.741 | 5/5  | 20    | 0     | 11.8  | 3      | 0.8    | 20    | 100       | 100         |
| 1.2 | 0.684 | 4/5  | 40    | 0     | 16.0  | 9      | 5.0    | 40    | 50        | 100         |
| 1.1 | 0.627 | 4/5  | 60    | 0     | 15.6  | 9      | 4.6    | 40    | 50        | 100         |
| 1.0 | 0.570 | 4/5  | 40    | 25    | 16.2  | 15     | 5.2    | 40    | 50        | 100         |
| 0.9 | 0.513 | 0/5  | 60    | 0     | 30.0  | 21     | 15.0   | 100   | 0         | —           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): align 100%, turn 67%, opening 65%, width 63%, blocked 60%, sideways 20%

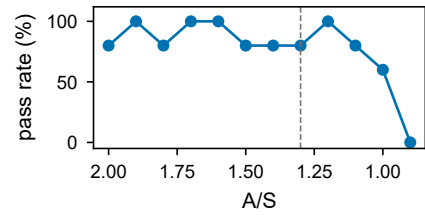
Actions: forward 72.4%, look\_down 8.7%, turn\_left 4.5%, right 3.5%, turn\_right 3.1%, left 2.6%

Strategy labels: FRONTAL 47%, ROT\_EARLY 40%, SIDEWAYS 8%, ROT\_MID 3%, ROT\_LATE 2%

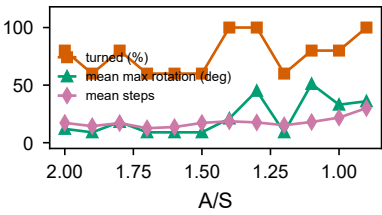
It turned in 50% of episodes (first turn at x = 2.5 m on average), and in 49% of the episodes at apertures that need no turn at all; its passes waste 5.3 steps over the optimum on average.

Inspecting its own body, compared inside each aperture: -40.0 percentage points on the pass rate (0 of 10 apertures positive).

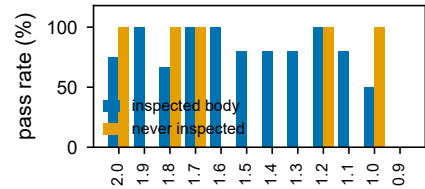
outcome vs aperture



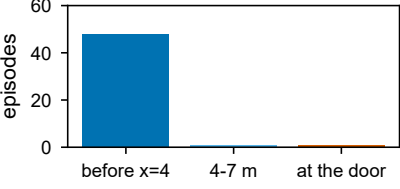
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.4 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 4/5  | 80    | 0     | 17.2  | 12     | 6.2    | 80    | 75        | 100         |
| 1.9 | 1.083 | 5/5  | 60    | 0     | 14.4  | 9      | 3.4    | 100   | 100       | -           |
| 1.8 | 1.026 | 4/5  | 80    | 0     | 17.0  | 18     | 6.0    | 60    | 67        | 100         |
| 1.7 | 0.969 | 5/5  | 60    | 0     | 12.6  | 9      | 1.6    | 20    | 100       | 100         |
| 1.6 | 0.912 | 5/5  | 60    | 0     | 13.6  | 9      | 2.6    | 100   | 100       | -           |
| 1.5 | 0.855 | 4/5  | 60    | 0     | 17.2  | 9      | 6.2    | 100   | 80        | -           |
| 1.4 | 0.798 | 4/5  | 100   | 0     | 18.6  | 21     | 7.6    | 100   | 80        | -           |
| 1.3 | 0.741 | 4/5  | 100   | 0     | 17.6  | 45     | 6.6    | 100   | 80        | -           |
| 1.2 | 0.684 | 5/5  | 60    | 0     | 15.2  | 9      | 4.2    | 80    | 100       | 100         |
| 1.1 | 0.627 | 4/5  | 80    | 25    | 18.0  | 51     | 7.0    | 100   | 80        | -           |
| 1.0 | 0.570 | 3/5  | 80    | 33    | 21.8  | 33     | 10.8   | 80    | 50        | 100         |
| 0.9 | 0.513 | 0/5  | 100   | 0     | 30.0  | 36     | 15.0   | 80    | 0         | 0           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): align 100%, opening 97%, turn 97%, width 93%, blocked 85%, sideways 37%, budget 5%

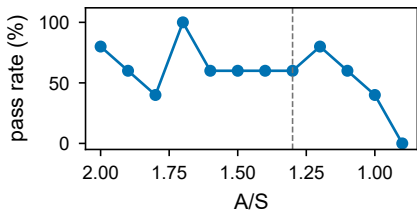
Actions: forward 68.0%, look\_down 7.4%, turn\_right 7.1%, turn\_left 5.6%, look\_left 4.1%, backward 2.9%

Strategy labels: ROT\_EARLY 73%, FRONTAL 23%, ROT\_LATE 2%, SIDEWAYS 2%

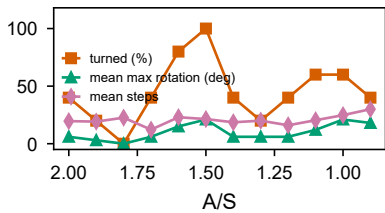
It turned in 77% of episodes (first turn at x = 1.8 m on average), and in 75% of the episodes at apertures that need no turn at all; its passes waste 6.4 steps over the optimum on average.

Inspecting its own body, compared inside each aperture: -18.1 percentage points on the pass rate (0 of 6 apertures positive).

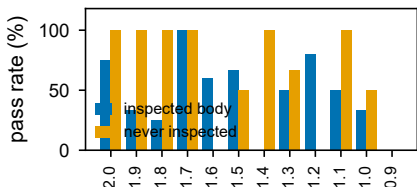
outcome vs aperture



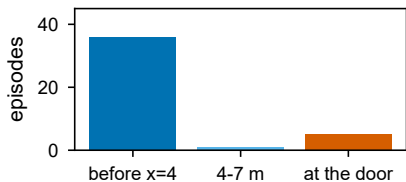
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.7 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

| A/S | width | pass | turn% | side% | steps | maxrot | excess | look% | pass look | pass nolook |
|-----|-------|------|-------|-------|-------|--------|--------|-------|-----------|-------------|
| 2.0 | 1.140 | 4/5  | 40    | 0     | 19.6  | 6      | 8.6    | 80    | 75        | 100         |
| 1.9 | 1.083 | 3/5  | 20    | 0     | 19.2  | 3      | 8.2    | 60    | 33        | 100         |
| 1.8 | 1.026 | 2/5  | 0     | 0     | 22.6  | 0      | 11.6   | 80    | 25        | 100         |
| 1.7 | 0.969 | 5/5  | 40    | 0     | 12.2  | 6      | 1.2    | 40    | 100       | 100         |
| 1.6 | 0.912 | 3/5  | 80    | 0     | 23.0  | 15     | 12.0   | 100   | 60        | —           |
| 1.5 | 0.855 | 3/5  | 100   | 0     | 21.4  | 21     | 10.4   | 60    | 67        | 50          |
| 1.4 | 0.798 | 3/5  | 40    | 0     | 18.6  | 6      | 7.6    | 40    | 0         | 100         |
| 1.3 | 0.741 | 3/5  | 20    | 0     | 20.0  | 6      | 9.0    | 40    | 50        | 67          |
| 1.2 | 0.684 | 4/5  | 40    | 0     | 16.0  | 6      | 5.0    | 100   | 80        | —           |
| 1.1 | 0.627 | 3/5  | 60    | 0     | 20.4  | 12     | 9.4    | 80    | 50        | 100         |
| 1.0 | 0.570 | 2/5  | 60    | 50    | 24.8  | 21     | 13.8   | 60    | 33        | 50          |
| 0.9 | 0.513 | 0/5  | 40    | 0     | 30.0  | 18     | 15.0   | 100   | 0         | —           |

Columns: pass|look / pass|nolook are this model's pass rate inside that aperture when it inspected its own body and when it did not; look% is how often it did.

Reasoning mentions (share of episodes): align 100%, opening 87%, width 83%, blocked 80%, turn 68%, sideways 32%, budget 3%

Actions: forward 70.3%, look\_down 9.4%, right 4.5%, left 3.8%, look\_left 3.3%, turn\_left 2.7%

Strategy labels: FRONTAL 43%, ROT\_EARLY 32%, SIDEWAYS 23%, ROT\_LATE 2%

It turned in 45% of episodes (first turn at x = 1.2 m on average), and in 45% of the episodes at apertures that need no turn at all; its passes waste 9.3 steps over the optimum on average.

Inspecting its own body, compared inside each aperture: -37.0 percentage points on the pass rate (1 of 9 apertures positive).